WINNING THE SPACE RACE

WITH DATA SCIENCE

Predicting Falcon 9 first-stage landing outcomes

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Outline

01
Problem + methods

02
Exploratory visual analysis

03
SQL results

04
Mapping + interactive analytics

05
Classification + conclusions

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Executive summary | Tested findings

WHY

The commercial objective is to understand which launch characteristics correlate with first-stage recovery, potentially informing assessments of booster reuse.

METHOD

Transcribed the 56-row public IBM dashboard sample; conducted ten SQL queries, six exploratory plots, geographic mapping, and cross-validated training of four classifiers.

MEASURED OUTCOMES

24 of 56 historical records indicate successful recovery (42.9%). A tuned linear SVM led 3-fold cross-validation (78.6%) and scored 78.6% on the 14-record holdout. The dataset is small.

DECISION IMPLICATION

Flight maturity and booster generation show relationships with landing outcome, but site and payload differences are confounded. These exploratory models are not operational forecasts.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Introduction | Business question and boundaries

BUSINESS PROBLEM

Falcon 9 reusability affects the economics of launch services. The analytical question: can historical mission characteristics help classify successful first-stage recovery?

LABEL

Class = 1 denotes a favorable historical landing outcome in the public dashboard sample; Class = 0 denotes the opposite. This must not be confused with overall mission success.

SCOPE

The executed evidence in this report is based on 56 public, historical dashboard rows. IBM also uses a separate 90-launch API dataset. Results across these datasets cannot be combined without reconciliation.

Methodology | End-to-end research design

1. **Source & transcribe public course CSV**
2. Validate 56 rows and target labels
3. SQL aggregation & EDA charts
4. Map site coordinates and outcomes
5. Create static dashboard views
6. Train four models, tune & test

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Data collection | SpaceX REST API workflow

PROCESS

GET /v4/launches/past → JSON normalization → join rocket, payload, core and launchpad records → choose Falcon 9 → export a rectangular analysis table.

WEB AND API SOURCES

API practice endpoint: api.spacexdata.com/v4/launches/past. The executed charts here use the historical dashboard CSV sourced from a publicly available course repository, not a newly issued live API request.

REPRODUCIBILITY LINK

Source dataset: github.com/Ahmed-Nezar/IBM-Data-Science-Capstone-SpaceX/blob/main/spacex_launch_dash.csv. A personal GitHub repository with the completed course notebooks remains outstanding.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Data collection | Wikipedia web scraping

EXTRACTION WORKFLOW

GET Wikipedia list of Falcon 9 and Falcon Heavy launches → BeautifulSoup parse HTML → inspect tables → extract launch identifiers, dates, sites and outcomes → standardize fields → dataframe.

QUALITY CONTROLS

Inspect table headings before parsing, clean footnote markers, remove incomplete or duplicated launch records, align recovery outcome terminology, and record source snapshots.

EVIDENCE STATUS

This describes the IBM course methodology. No original web-scraping notebook output or completed personal notebook URL was supplied; this section is not represented as an executed scrape.

Data wrangling | Steps actually applied

NORMALIZATION

Created named columns for flight number, site, landing Class, payload mass (kg) and booster category; validated 56 distinct flight identifiers and binary labels.

DATA CLEANING

Kept source-reported payload values including zeros; did not impute unknown outcomes or rewrite labels. Exported the complete 56-row analysis CSV and a SQLite database.

MODEL FEATURES

Used flight number and payload as continuous predictors, and one-hot encoded launch site and booster category. Split 75% training/25% testing with stratification and fixed random state.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

EDA methodology | Chart selection

SCATTER PLOTS

Flight number or payload mass by launch site and by booster group, colored by Class. This exposes separation, overlap, launch cadence, and possible outliers.

BAR CHARTS

Mean of Class grouped by site and booster generation represents percent favorable landing outcome. Each label includes its sample size to highlight instability.

TEMPORAL PROXY

The 56-row Dash dataset lacks launch dates. Flight-number quartiles serve only as a chronological proxy; the figure is explicitly not called a calendar-year trend.

SQL methodology | Questions addressed

QUERY CLASSES

GROUP BY site / booster; COUNT flights; AVG(Class) as a landing proportion; payload averages/maxima; CASE payload bands; SELECT … LIMIT; ORDER BY rankings.

SYSTEM

SQLite database built from the same 56-row historical CSV to keep query totals consistent with the visualizations. Ten query outputs are reproduced below.

IMPORTANT LIMIT

This CSV has recovery Class, not separate mission outcomes or the same NASA-booster labels as the IBM SQL lab. Rather than invent those missing observations, analyses are limited to available fields.

Interactive mapping and dashboard methodology

FOLIUM

Map site coordinates and scale CircleMarkers by observed flight count. Marker tooltips include total launches and successful outcomes; the delivered HTML remains interactive.

DASH ANALYTICS

Site dropdown, payload filter, outcome distribution and payload-vs-flight scatter views are constructed from the same historical sample. Static exports appear in this report.

NOTES

A geographic coordinate figure is a genuine generated plot but not a browser screenshot. Do not claim local infrastructure-proximity distances without a geospatial layer.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Predictive analysis methodology

ESTIMATORS

Logistic regression, support vector machine, decision tree, and k-nearest neighbors trained on identical splits; train count 42 / test count 14.

TUNING

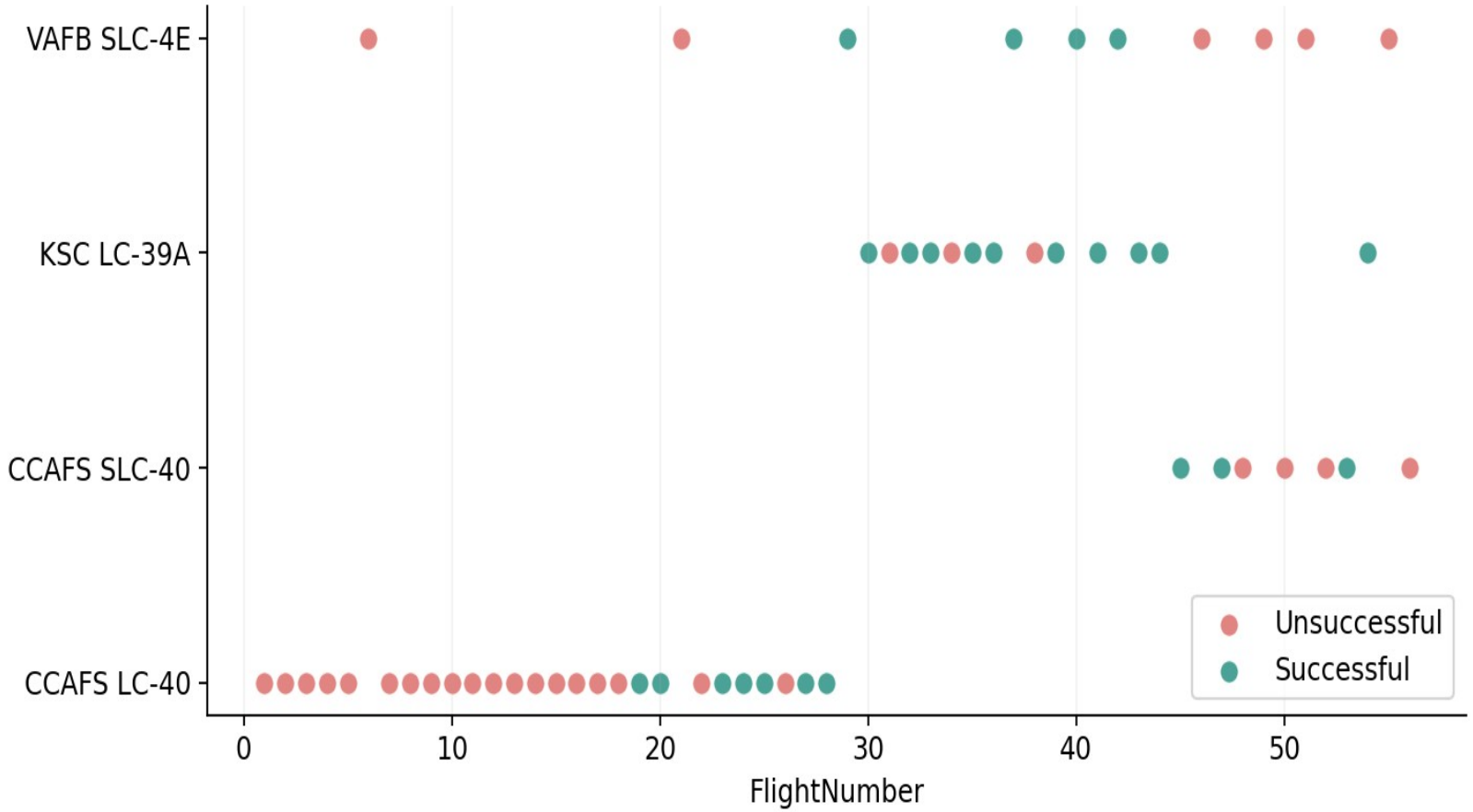
Stratified 3-fold cross-validation on training observations and parameter-grid search. Optimize mean validation accuracy, then report holdout test accuracy separately.

EVALUATION

Compare CV and test accuracy, select the winning estimator using cross-validation, inspect its confusion matrix, and acknowledge high sampling variance on fourteen test observations.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Flight number vs. launch site



WHAT THE RESULT SHOWS

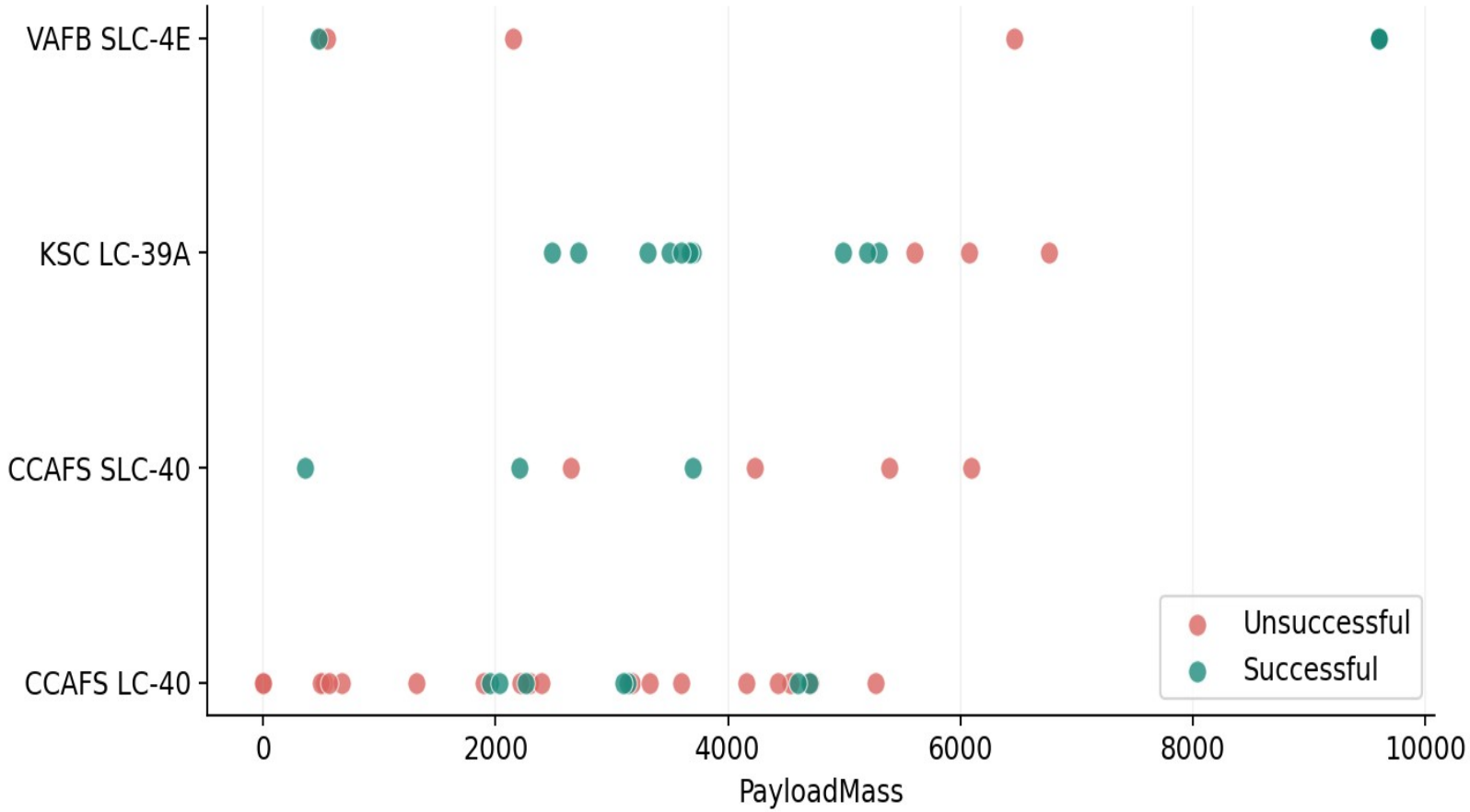
Recovery-positive launches concentrate more in later flight numbers, but early site-specific samples are limited.

HOW TO READ IT

Each dot = one flight; blue/green denotes Class 1 and red denotes Class 0.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Payload mass vs. launch site



WHAT THE RESULT SHOWS

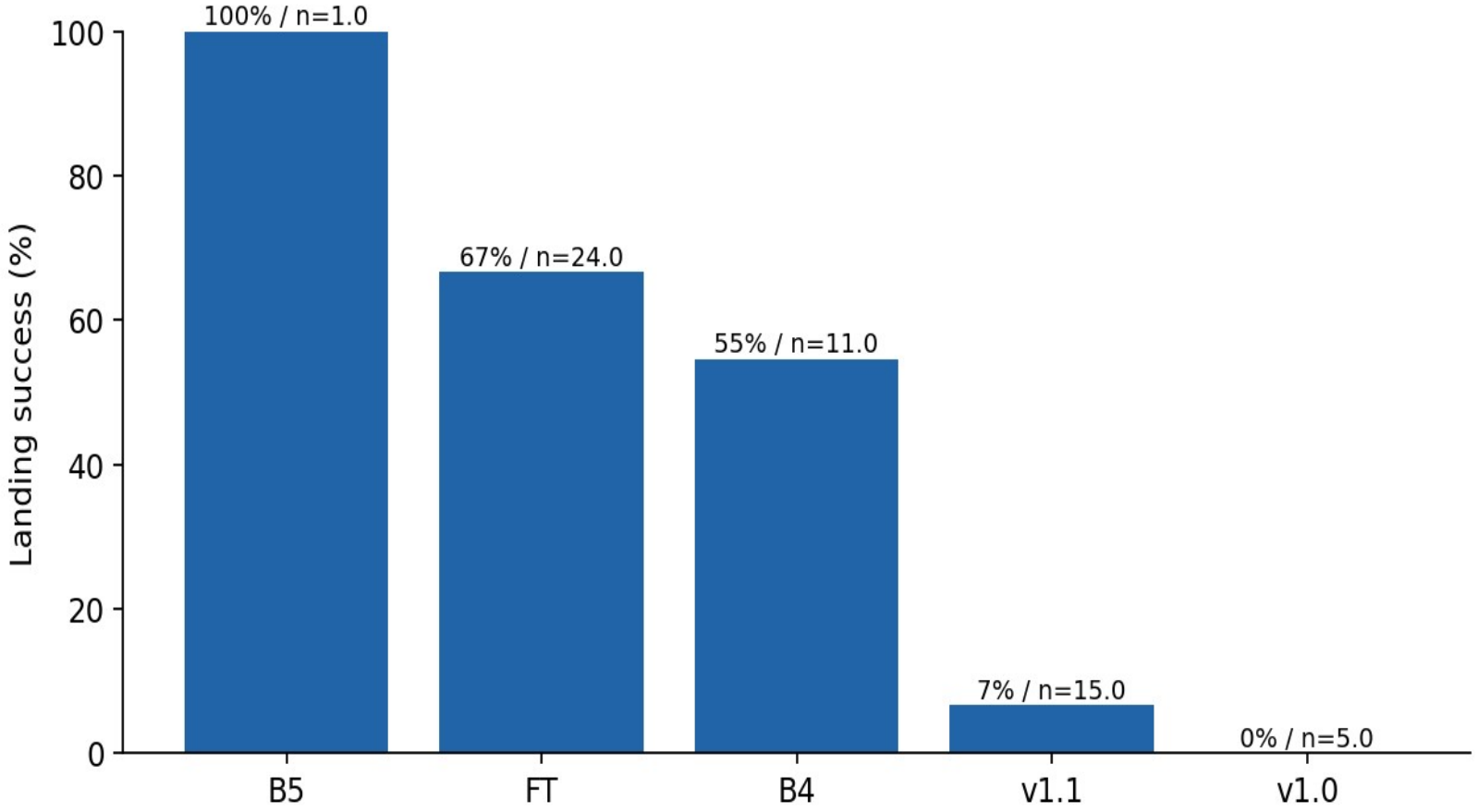
Payload distributions overlap across sites; a site effect could reflect mission profile and booster generation.

HOW TO READ IT

Position = mass/site, color = outcome. Zero-valued payloads are retained as in the source.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Landing success by booster generation



WHAT THE RESULT SHOWS

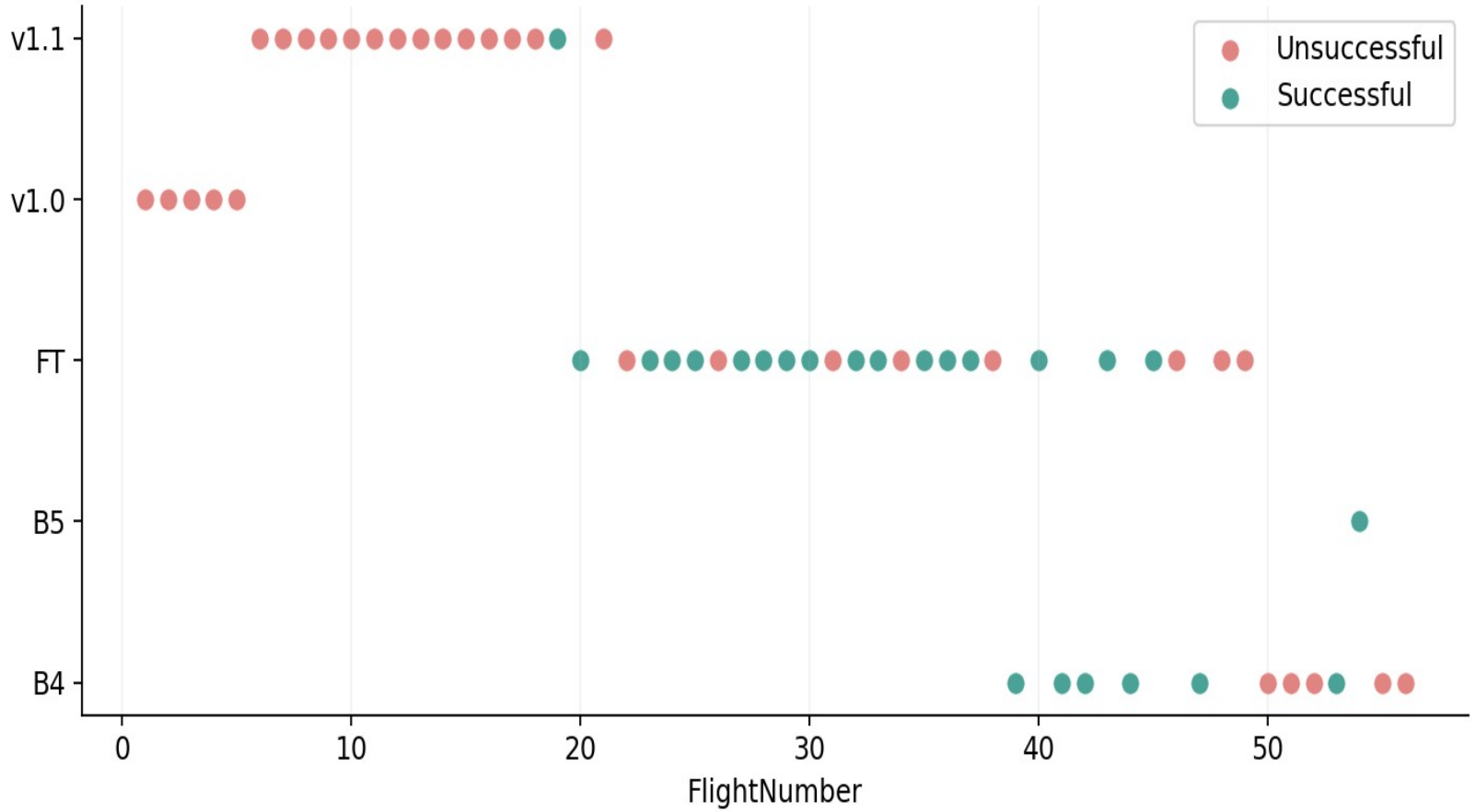
Later booster categories have higher recovery proportions in this selected sample; small group sizes limit generalization.

HOW TO READ IT

Bars are class means, shown as percentages and annotated with group sample sizes.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Flight number vs. booster generation



WHAT THE RESULT SHOWS

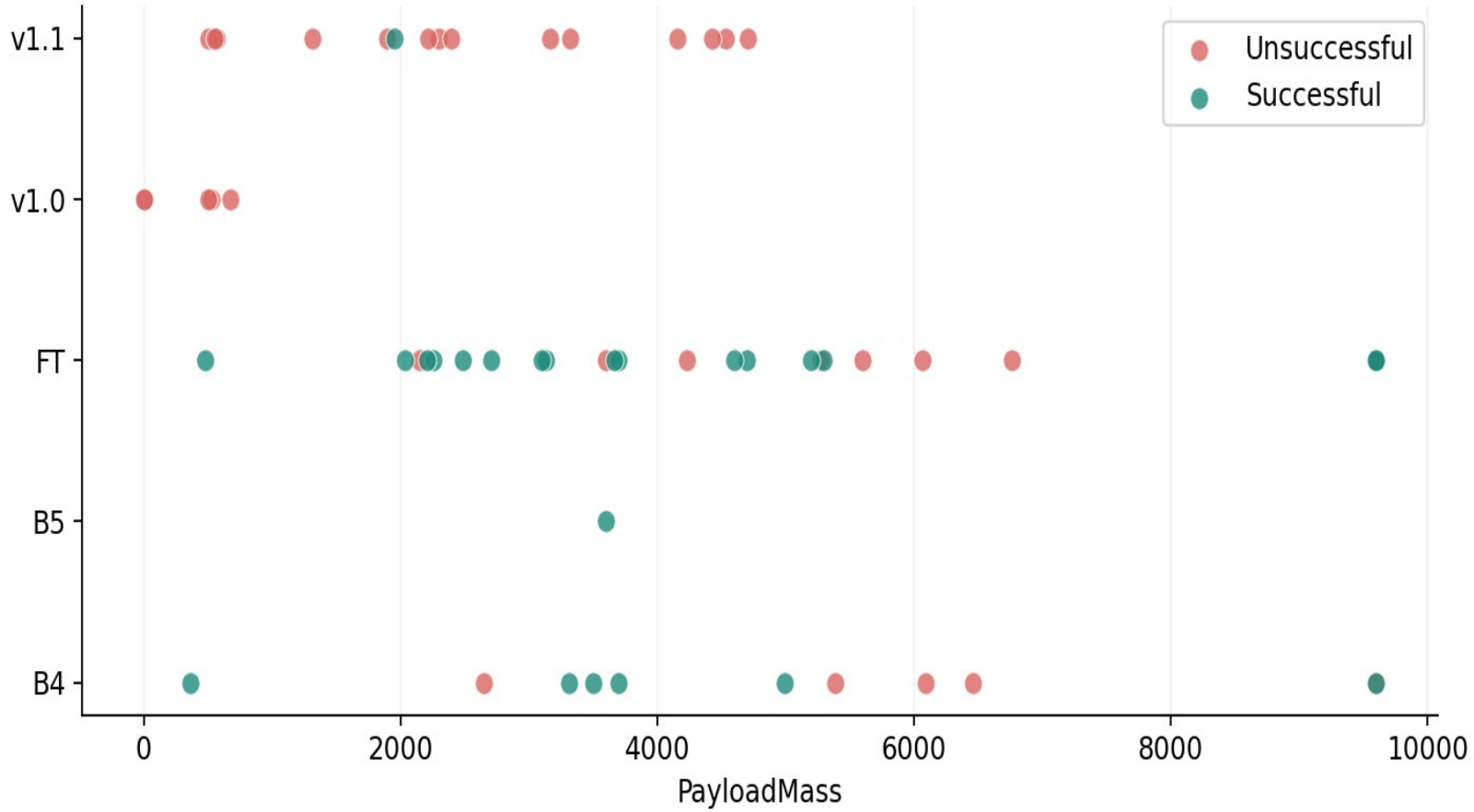
Booster generation changes over flight history, making a simple maturity narrative potentially confounded.

HOW TO READ IT

Scatterplot of actual rows with one point per historical flight.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Payload mass vs. booster generation



WHAT THE RESULT SHOWS

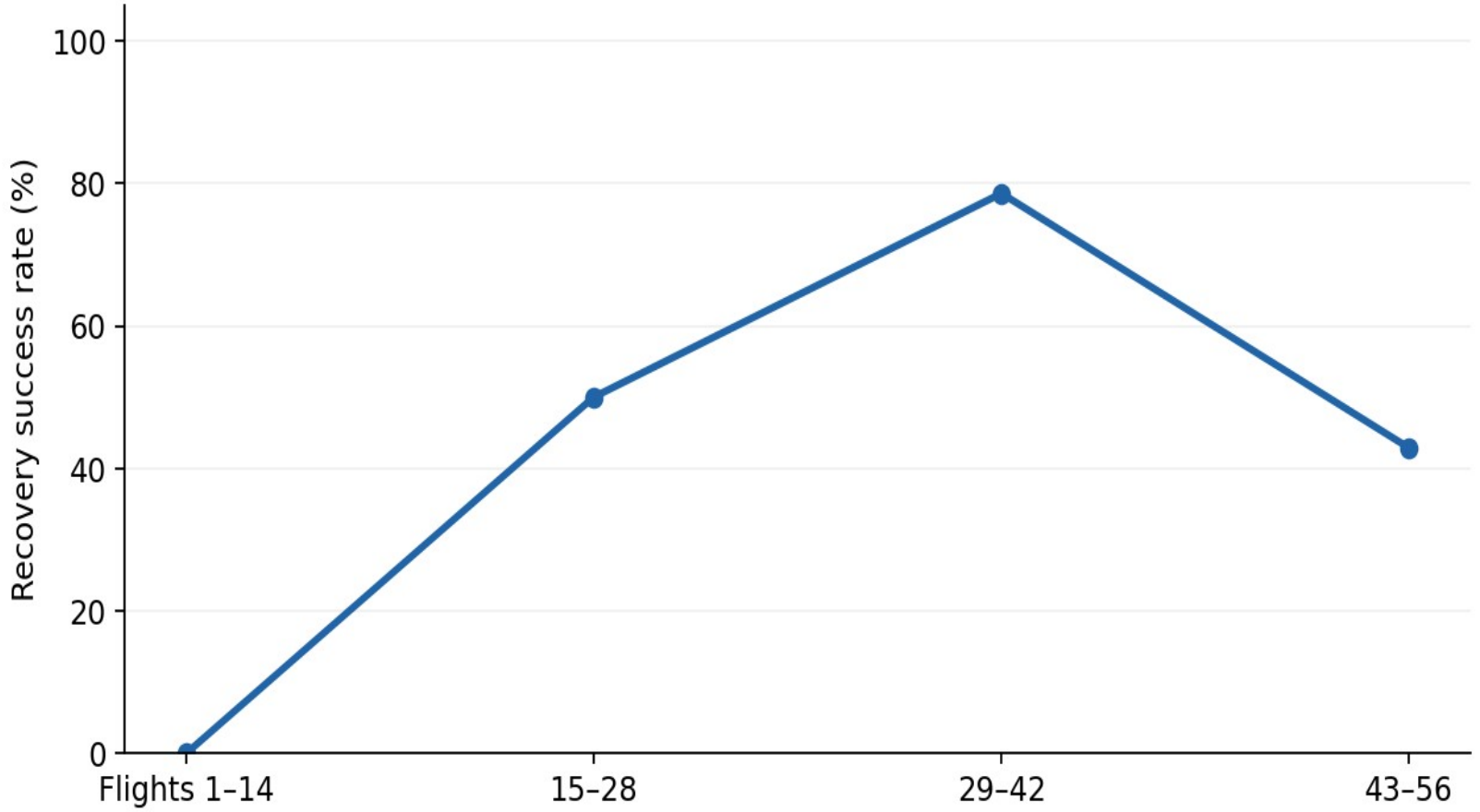
The range of payload masses differs across generations, so unadjusted recovery comparisons may be misleading.

HOW TO READ IT

This chart is categorical-by-continuous; class labels are the two point colors.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Flight progression vs. landing success



WHAT THE RESULT SHOWS

Average historical recovery generally improves across successive flight-number groups in this extract.

HOW TO READ IT

Four flight-number groups are used. NOT a calendar-year trend: the 56-row file contains no dates.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 01 | Launch sites and flight counts

```
SELECT LaunchSite, COUNT(*) AS launches FROM SPACEXTBL GROUP BY LaunchSite ORDER BY launches DESC
```

LaunchSite	launches
CCAFS LC-40	26
KSC LC-39A	13
VAFB SLC-4E	10
CCAFS SLC-40	7

INTERPRETATION

Two Cape Canaveral naming variants appear in the source and are kept separate to preserve provenance.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 02 | Landing success by site

```
SELECT LaunchSite, COUNT(*) AS n, SUM(Class) AS successful, ROUND(100.0*AVG(Class),1) AS success_pct FROM SPACEXTBL GROUP BY LaunchSite ORDER BY success_pct DESC
```

LaunchSite	n	successful	success_pct
KSC LC-39A	13	10	76.9
CCAFS SLC-40	7	3	42.9
VAFB SLC-4E	10	4	40.0
CCAFS LC-40	26	7	26.9

INTERPRETATION

Success percentage is AVG(Class) × 100, not mission completion or orbital-insertion success.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 03 | Payload by booster generation

```
SELECT BoosterCategory,COUNT(*) AS flights,ROUND(AVG(PayloadMass),0) AS avg_payload_kg,ROUND(MAX(PayloadMass),0) AS
max_payload_kg FROM SPACEXTBL GROUP BY BoosterCategory ORDER BY flights DESC
```

BoosterCategory	flights	avg_payload_kg	max_payload_kg
FT	24	4502.0	9600.0
v1.1	15	2535.0	4707.0
B4	11	5058.0	9600.0
v1.0	5	340.0	677.0
B5	1	3600.0	3600.0

INTERPRETATION

Comparing mean mass by booster group provides context before interpreting landing differences.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 04 | Successful vs unsuccessful

```
SELECT Class,COUNT(*) AS launches FROM SPACEXTBL GROUP BY Class ORDER BY Class DESC
```

Class	launches
1	24
0	32

INTERPRETATION

The dataset contains 24 Class=1 and 32 Class=0 records (42.9% favorable recovery).

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 05 | Heaviest payload missions

```
SELECT FlightNumber,LaunchSite,PayloadMass,Class FROM SPACEXTBL ORDER BY PayloadMass DESC LIMIT 5
```

FlightNumber	LaunchSite	PayloadMass	Class
29	VAFB SLC-4E	9600.0	1
37	VAFB SLC-4E	9600.0	1
42	VAFB SLC-4E	9600.0	1
46	VAFB SLC-4E	9600.0	0
51	VAFB SLC-4E	9600.0	0

INTERPRETATION

This ranking inspects unusually heavy payload records and whether their recovery label was favorable.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 06 | Cape Canaveral example rows

```
SELECT FlightNumber, LaunchSite, Class FROM SPACEXTBL WHERE LaunchSite LIKE 'CCA%' ORDER BY FlightNumber LIMIT 5
```

FlightNumber	LaunchSite	Class
1	CCAFS LC-40	0
2	CCAFS LC-40	0
3	CCAFS LC-40	0
4	CCAFS LC-40	0
5	CCAFS LC-40	0

INTERPRETATION

A simple prefix filter demonstrates SQL string matching and sample inspection.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 07 | Booster recovery-rate ranking

```
SELECT BoosterCategory, COUNT(*) AS flights, SUM(Class) AS successful, ROUND(100.0*AVG(Class),1) AS success_pct FROM SPACEXTBL GROUP BY BoosterCategory ORDER BY success_pct DESC
```

BoosterCategory	flights	successful	success_pct
B5	1	1	100.0
FT	24	16	66.7
B4	11	6	54.5
v1.1	15	1	6.7
v1.0	5	0	0.0

INTERPRETATION

A high rate with a small denominator should not be treated as evidence of robust superiority.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 08 | Payload-band success

```
SELECT CASE WHEN PayloadMass < 2000 THEN '<2,000 kg' WHEN PayloadMass < 5000 THEN '2,000–4,999 kg' ELSE '5,000+ kg' END AS payload_band, COUNT(*) AS flights, ROUND(100.0*AVG(Class),1) AS success_pct FROM SPACEXTBL GROUP BY payload_band ORDER BY flights DESC
```

payload_band	flights	success_pct	
"2	000–4	999 kg"	29
"5	000+ kg"	14	35.7
"<2	000 kg"	13	23.1

INTERPRETATION

The payload strata offer descriptive segmentation, not causal impact estimates.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 09 | Maximum-payload flights

```
SELECT FlightNumber, LaunchSite, BoosterCategory, PayloadMass FROM SPACEXTBL WHERE PayloadMass = (SELECT MAX(PayloadMass) FROM SPACEXTBL)
```

FlightNumber	LaunchSite	BoosterCategory	PayloadMass
29	VAFB SLC-4E	FT	9600.0
37	VAFB SLC-4E	FT	9600.0
42	VAFB SLC-4E	B4	9600.0
46	VAFB SLC-4E	FT	9600.0
51	VAFB SLC-4E	B4	9600.0

INTERPRETATION

Multiple launches can tie for the maximum mass; the query returns all matching cases.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

SQL 10 | Launch-site rank by successes

```
SELECT LaunchSite, COUNT(*) AS flights, SUM(Class) AS successes FROM SPACEXTBL GROUP BY LaunchSite ORDER BY successes DESC
```

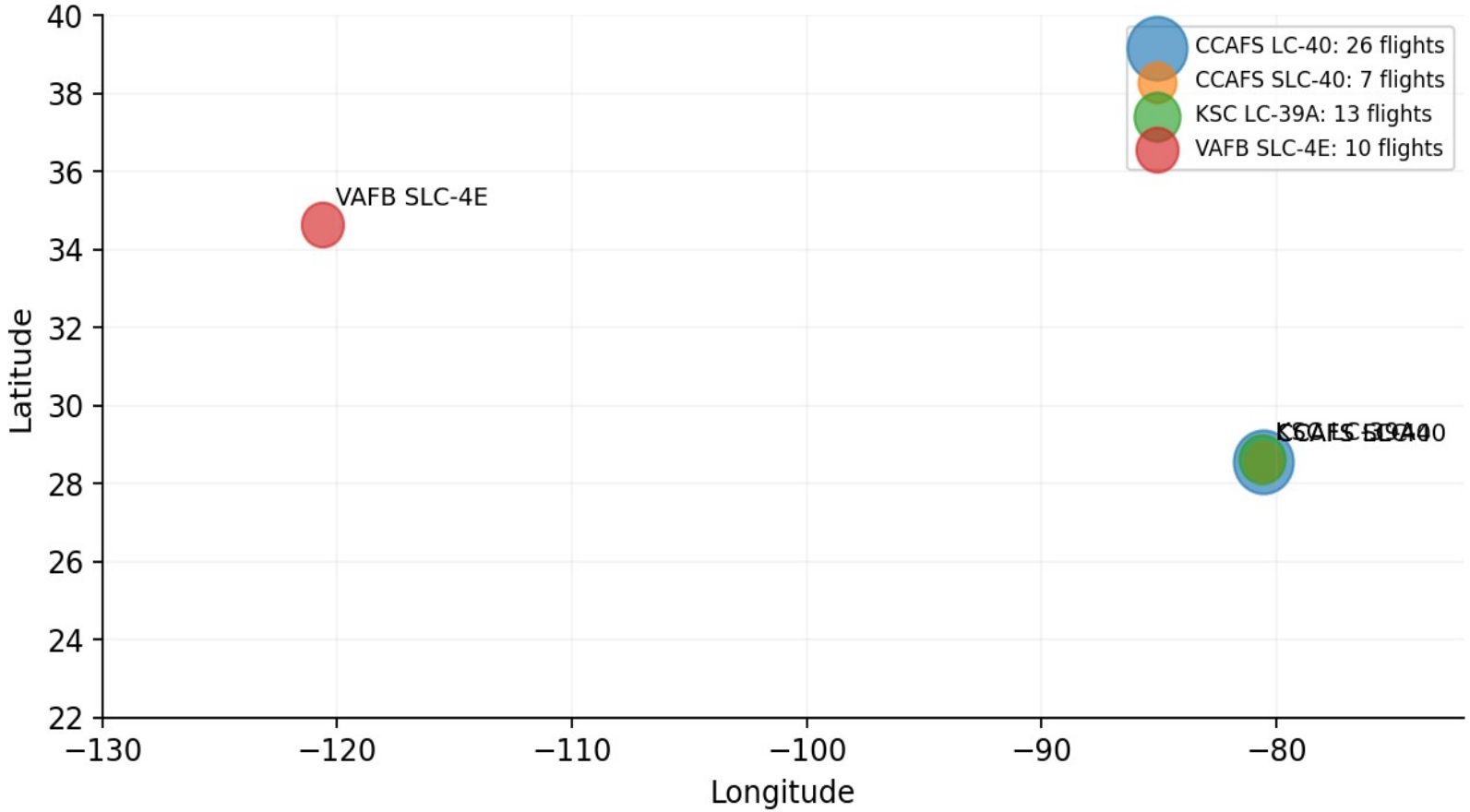
LaunchSite	flights	successes
KSC LC-39A	13	10
CCAFS LC-40	26	7
VAFB SLC-4E	10	4
CCAFS SLC-40	7	3

INTERPRETATION

Number of recoveries reflects both underlying success rate and how frequently each site was used.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Folium 01 | Geographic launch-site markers



WHAT THE RESULT SHOWS

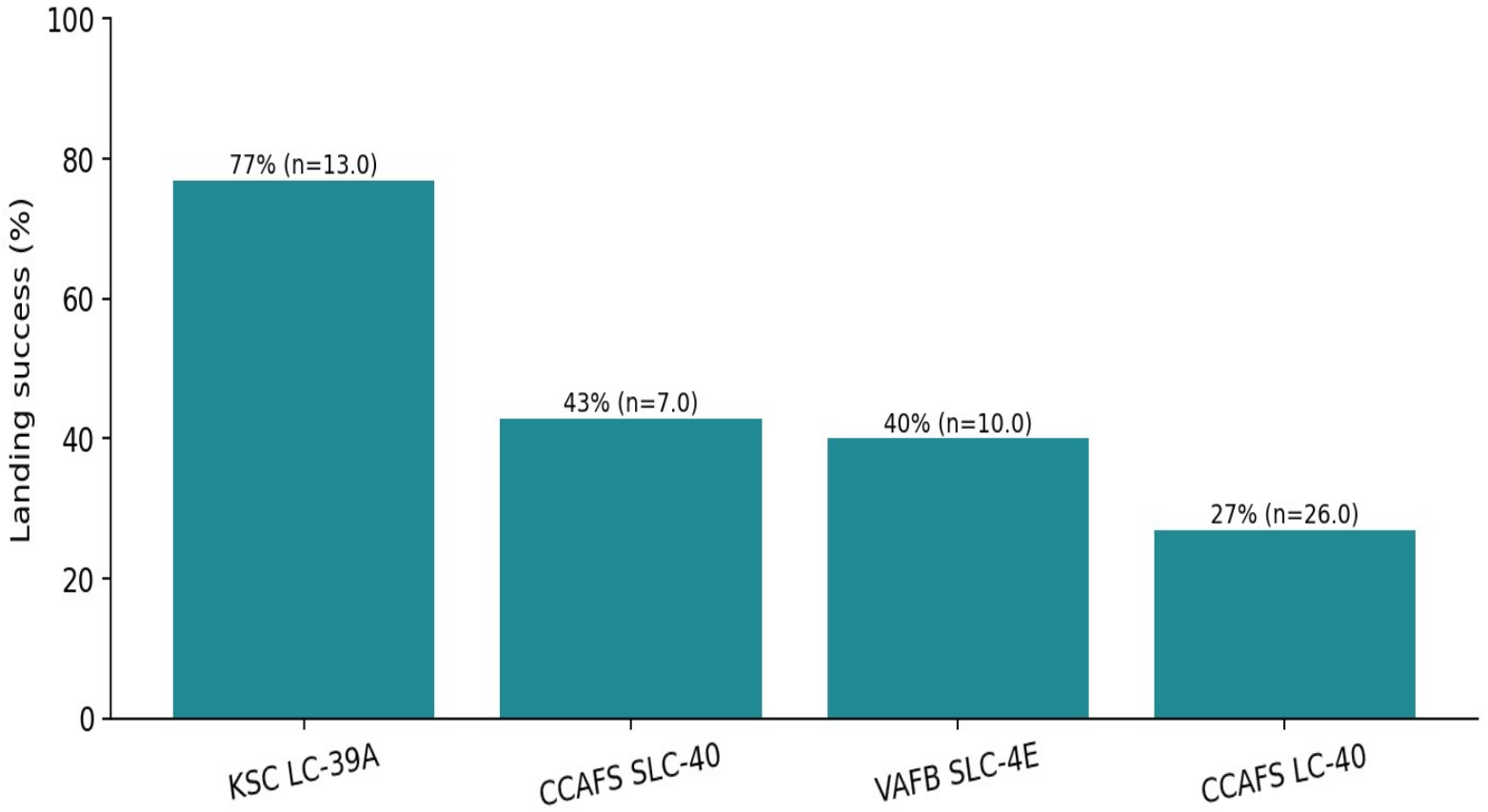
Three distinct geographic positions are visible; two Cape Canaveral labels share one coordinate.

HOW TO READ IT

Marker area indicates sample count. Interactive Folium HTML is included in supporting assets.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Folium 02 | Outcome-coded launch locations



MAP INTERPRETATION

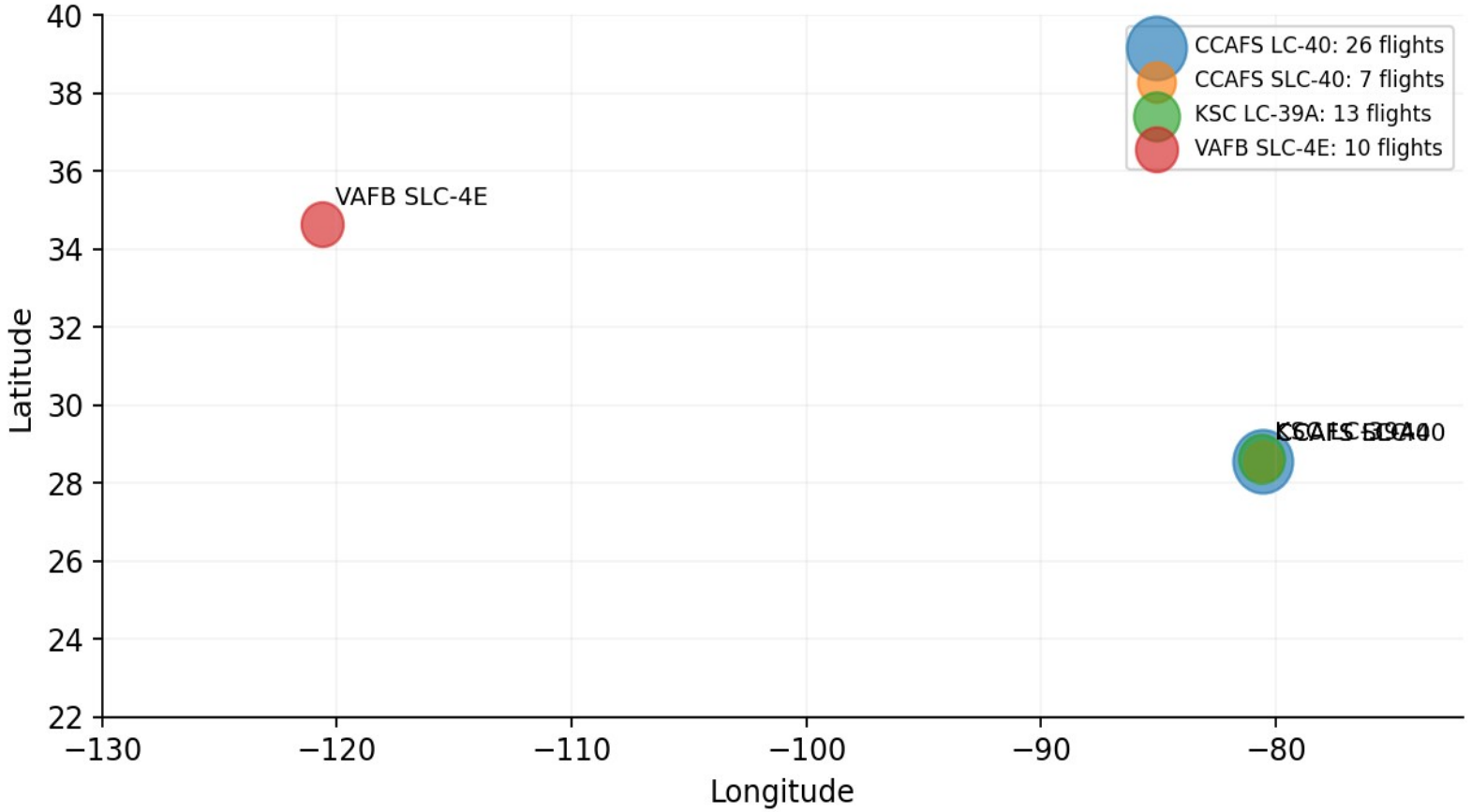
Site-specific percentages can be added as tooltip labels in Folium. Different denominators make direct visual comparisons uncertain.

OUTPUT EVIDENCE

The companion folium_site_map.html holds actual clickable site markers with success totals. This slide combines a quantitative companion plot rather than pretending a screenshot.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Folium 03 | Proximity and infrastructure limits



COASTAL LOCATION

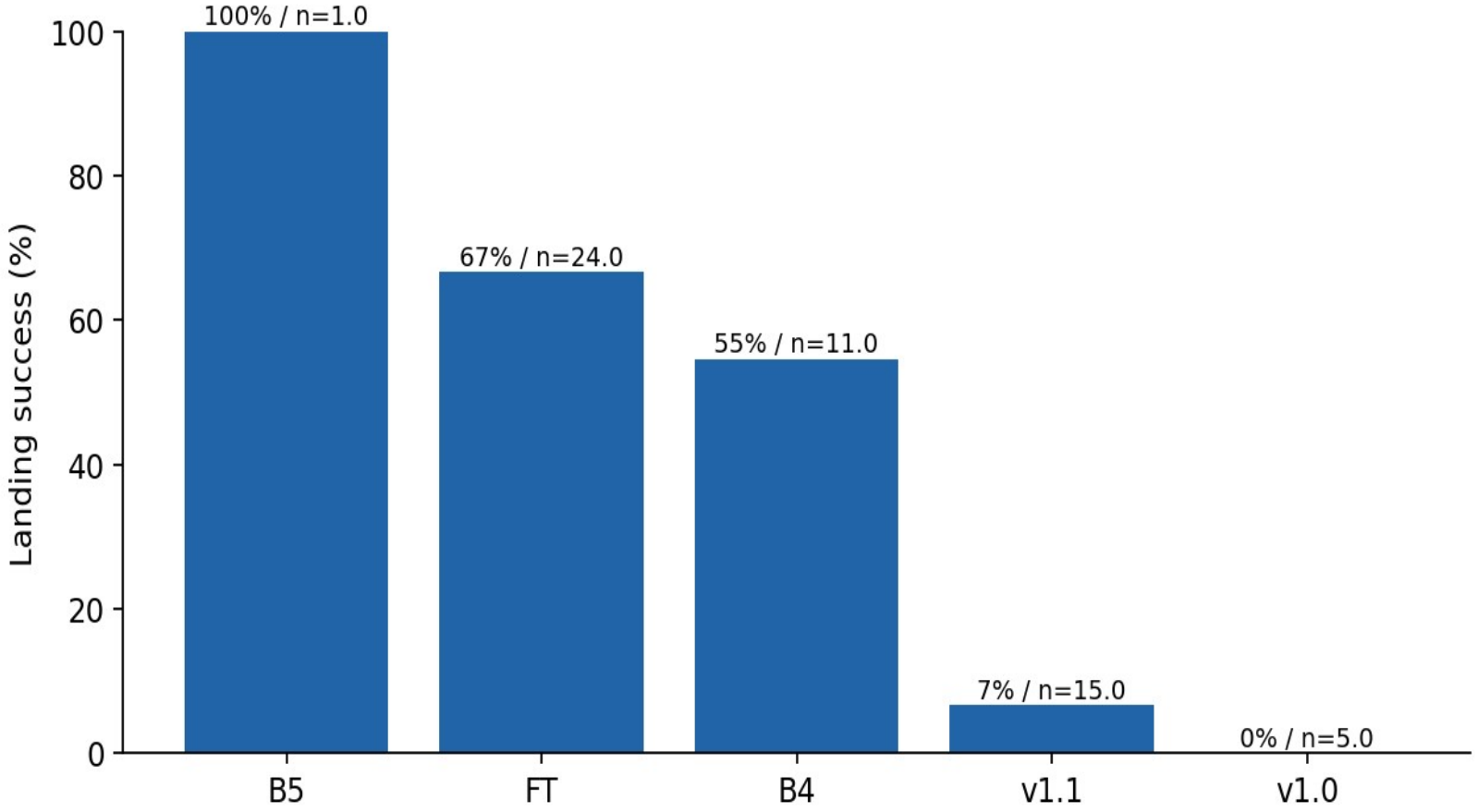
Florida and California pads are placed near coastlines. The supplied data contain coordinates only.

DISTANCE REQUIREMENTS

Distances to railways, highways, or coastlines cannot be verified without geometry or map layers; no invented kilometer measurements are shown.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Dashboard 01 | Overall outcome proportions



WHAT THE RESULT SHOWS

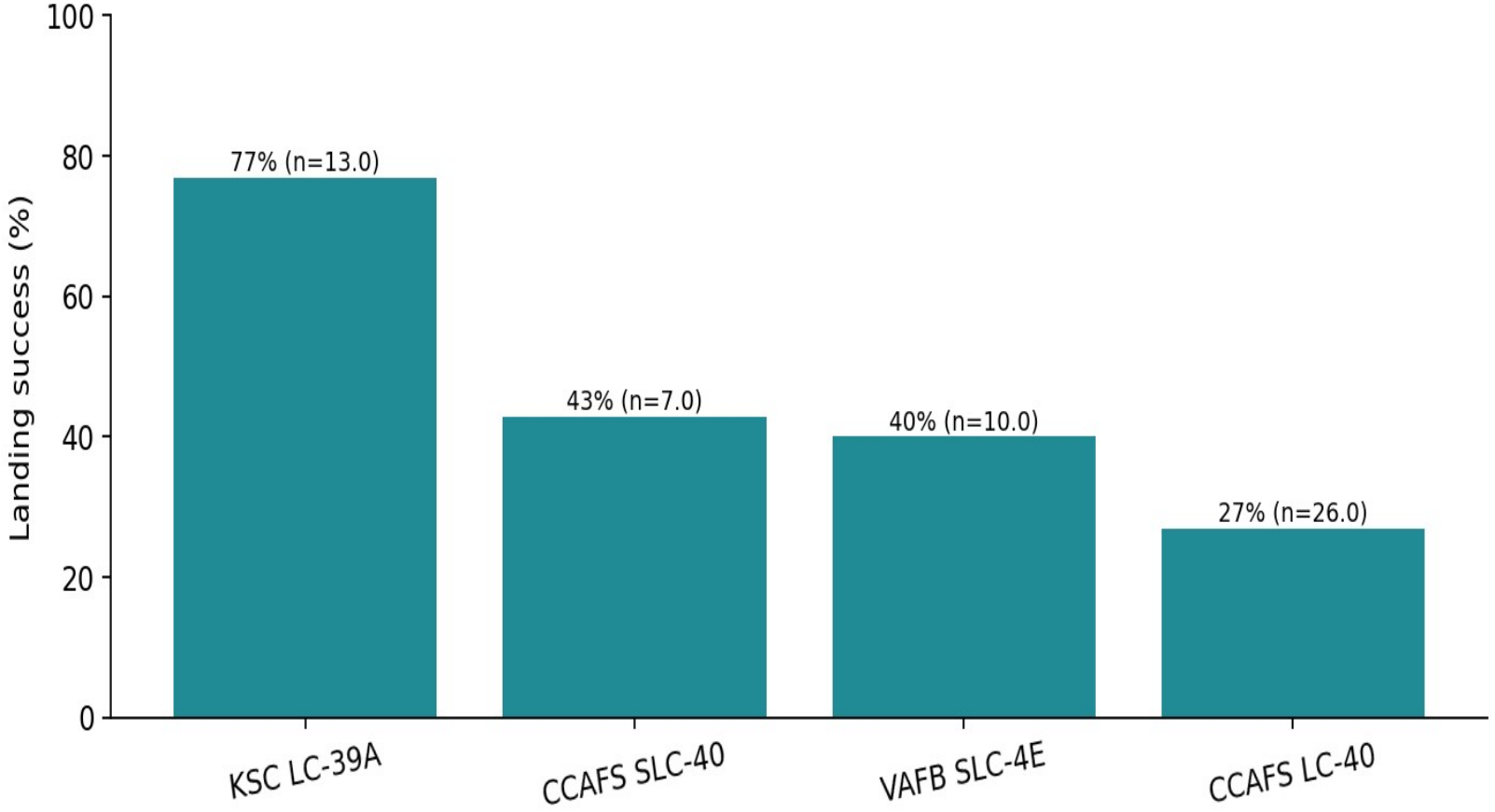
Historical recoveries are not evenly distributed across booster categories.

HOW TO READ IT

The implemented dashboard should allow users to choose all sites or an individual pad.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Dashboard 02 | Compare launch sites



WHAT THE RESULT SHOWS

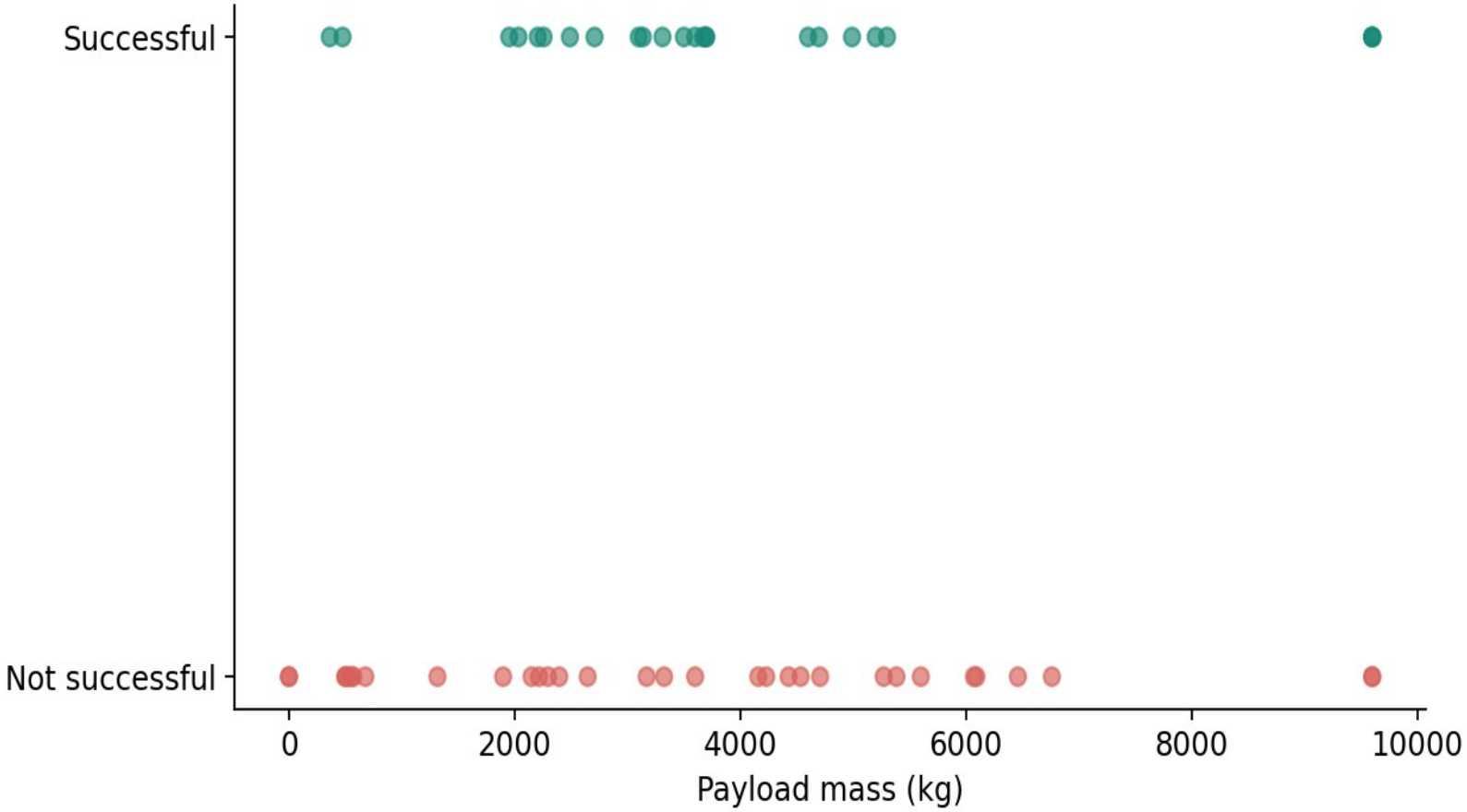
A site-dropdown view isolates the landing rates and denominator at each site.

HOW TO READ IT

Use the exact selected subset when comparing outcomes; do not infer a site causal effect.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Dashboard 03 | Payload range and outcome



WHAT THE RESULT SHOWS

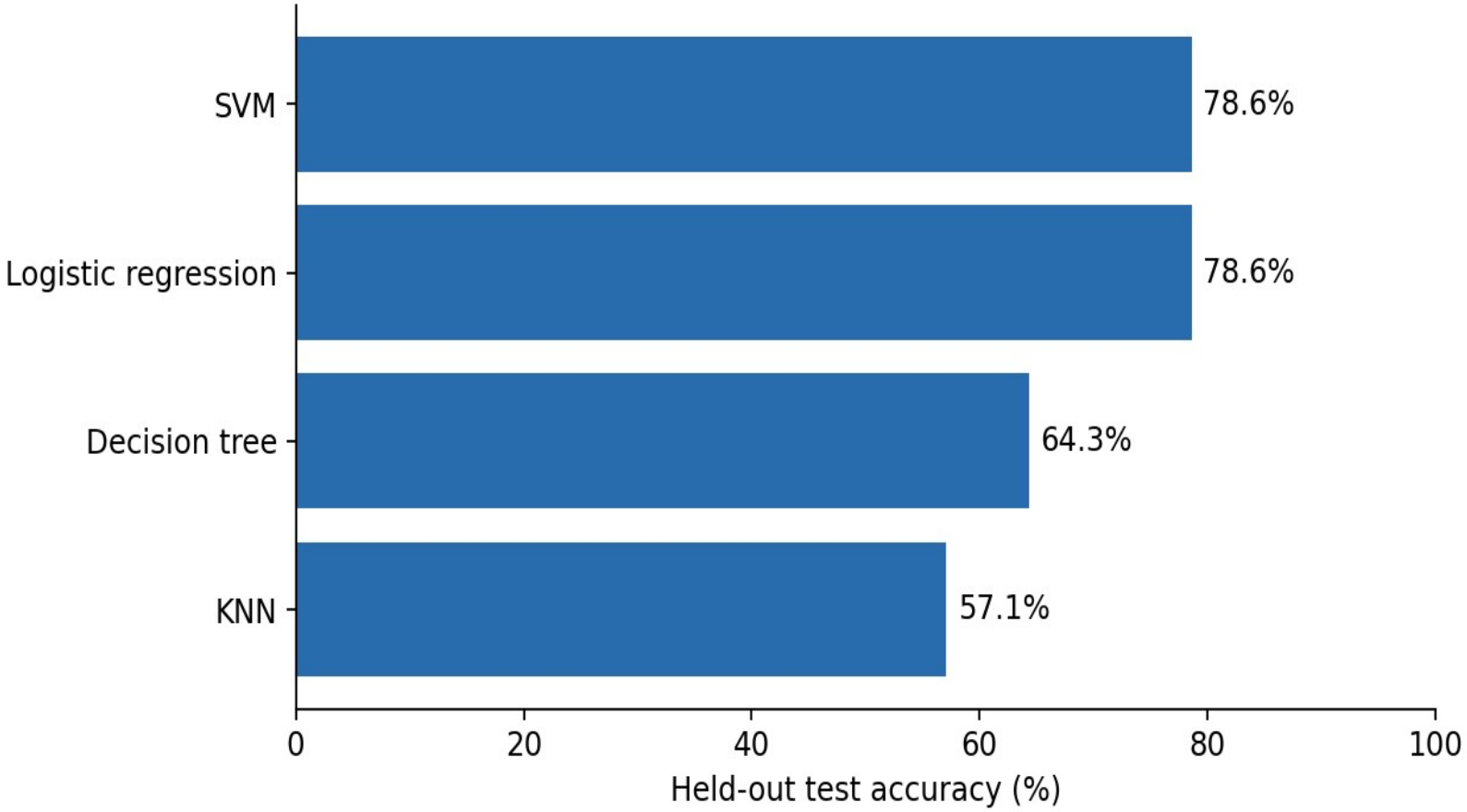
A payload range slider can isolate different regions of the observed recovery labels.

HOW TO READ IT

This static dashboard evidence comes from generated data plots, not an executed browser screenshot.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Machine learning | Four-model comparison



WHAT THE RESULT SHOWS

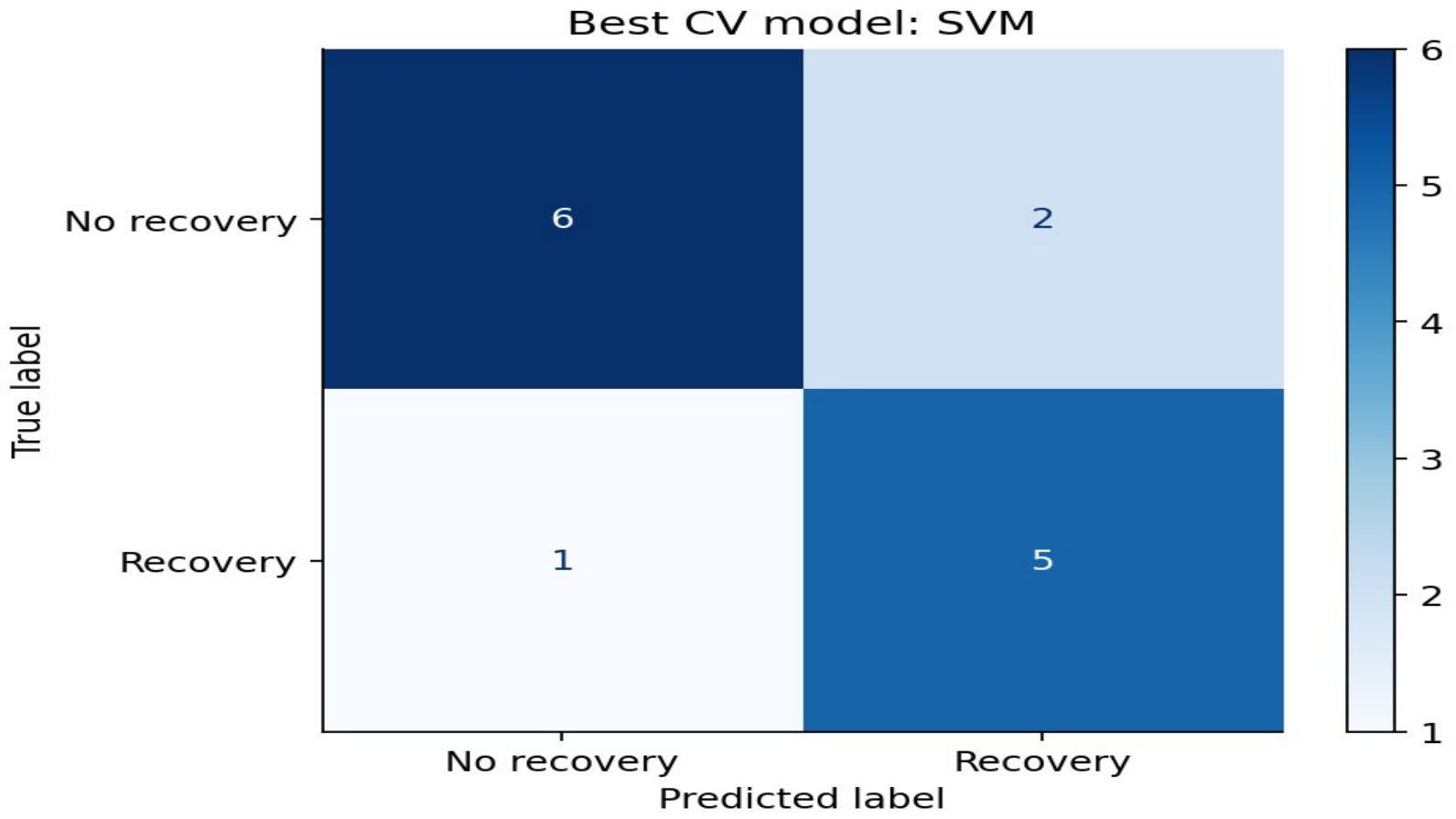
SVM has the highest cross-validation accuracy (78.6%); logistic regression ties its held-out score.

HOW TO READ IT

42 training rows; 14 test rows; 3-fold CV. These are actual locally computed sample results.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Best model | Confusion matrix



WHAT THE RESULT SHOWS

The confusion matrix shows which Class labels the tuned SVM misclassified. False positive and false negative implications differ.

HOW TO READ IT

Use this to supplement accuracy, which can be misleading in a small dataset.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Predictive performance | Validation vs test

Model	CV accuracy	Test accuracy
SVM	78.6%	78.6%
Logistic regression	73.8%	78.6%
Decision tree	73.8%	64.3%
KNN	73.8%	57.1%

MODEL CHOICE

Select SVM based on highest validation accuracy, not its test score. Its holdout result is 11 correct predictions in 14 cases (78.6%).

REPLICATION

Re-run with the official 90-row IBM dataset and its exact notebook pipeline before comparing to your course quiz results.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Conclusion | Insights and business implications

01 RECOVERY IS NOT RANDOM

The observed recovery rate is 24/56 = 42.9%, but this historical snapshot is not representative of current Falcon 9 operations.

02 BOoster MATURITY MATTERS

Flight-number and booster-category patterns support investigating learning and hardware maturity, while leaving substantial confounding.

03 CLASSIFIERS CAN BE INFORMATIVE

The tuned SVM achieved 78.6% cross-validation and held-out accuracy on this small sample. The test score has large uncertainty.

04 NEXT DECISION

Validate a later, larger dataset with date-based holdouts and more mission covariates before any procurement, cost, or risk decisions.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Appendix | Sources, deliverables and integrity

PUBLIC DATA SOURCE

github.com/Ahmed-Nezar/IBM-Data-Science-Capstone-SpaceX/blob/main/spacex_launch_dash.csv • Historical 56-row training visualization sample (transcribed for local reproducibility).

GENERATED OUTPUTS

CSV, SQL database and query tables, eleven generated plots, four fitted models with parameters, model accuracies, a confusion matrix, and an interactive Folium HTML.

REMAINING PERSONAL-SUBMISSION REQUIREMENTS

No personal GitHub account or completed original IBM API, scraper, SQL, Folium, or Dash notebooks were provided. Upload your own repository and completed notebooks to meet rubric item 1.1 and its source-link criteria.

Submission evidence | Executed vs outstanding

EXECUTED IN THIS PACKAGE

A complete local notebook with saved output cells; ten SQLite query tables; generated EDA charts; fitted classifier comparison; confusion matrix and interactive Folium HTML.

STARTER LABS RECEIVED

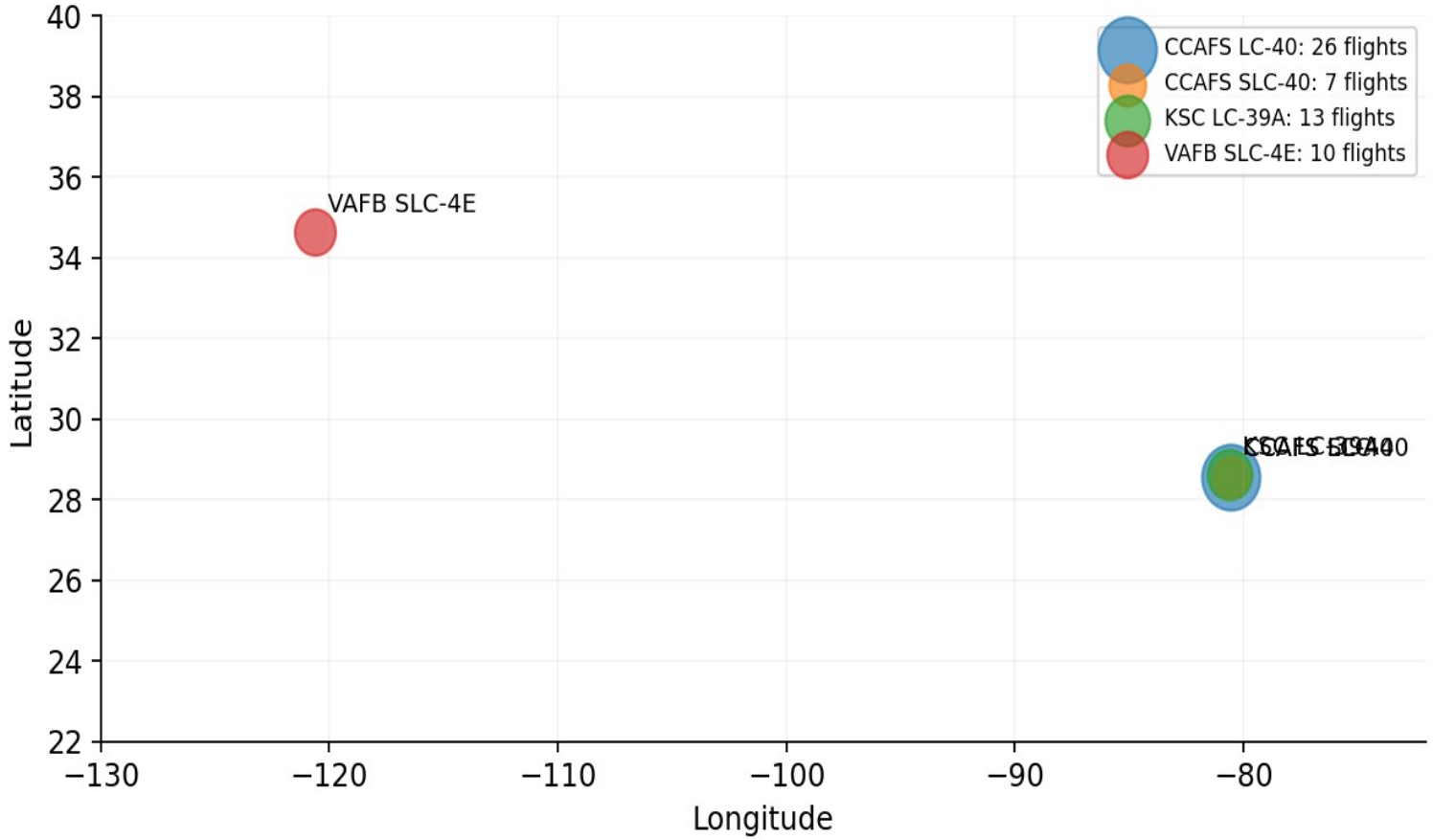
SpaceX API, Wikipedia scraping, wrangling, visualization, Folium, and machine-learning notebooks were supplied with blank output cells. The Dash lab was supplied as instructions.

STILL NEEDED FOR FULL IBM RUBRIC

The source-specific 90-row API/orbit/year results; verified Dash browser screenshots; precise geographic infrastructure proximity; completed original lab notebooks; and a GitHub link owned by the learner.

Dataset: historical IBM 56-row dashboard sample • distinct from 90-row API lab

Folium evidence | Real launch-site location data



WHAT THE MAP SUPPORTS

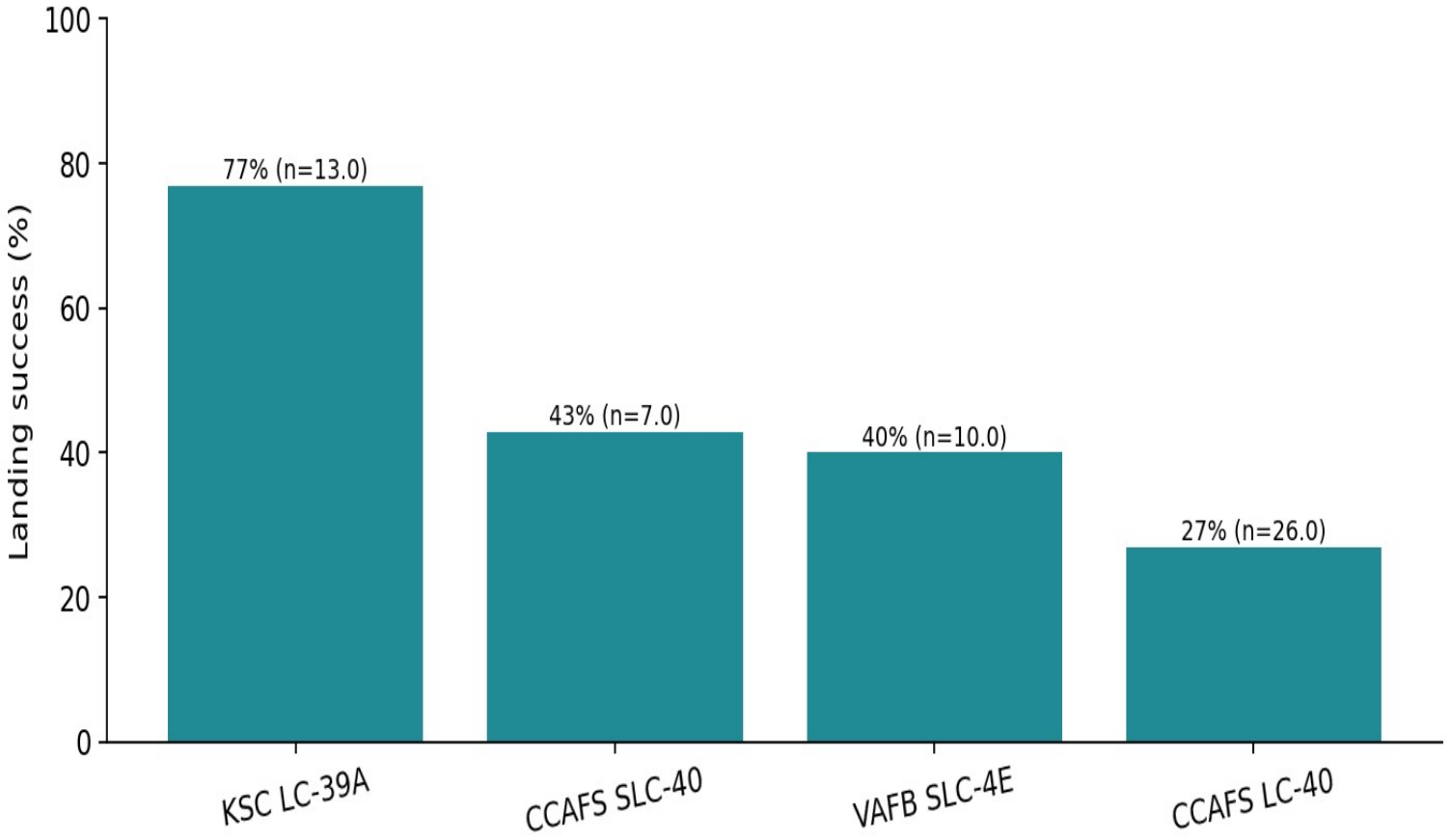
Actual plotted launch-pad coordinates place sites in Florida and California; CircleMarker size represents flight count. The companion interactive Folium HTML includes clickable site-level data.

REPRODUCIBILITY

folium_site_map.html is provided. This geographic graphic is generated from the site coordinates, rather than a captured Folium browser view.

Source: bundled launch-site coordinates and 56-record course dashboard extract.

Geography | Outcome counts at each launch pad



OBSERVED OUTCOMES

KSC LC-39A: 10/13 (76.9%)

CCAFS LC-40: 7/26 (26.9%)

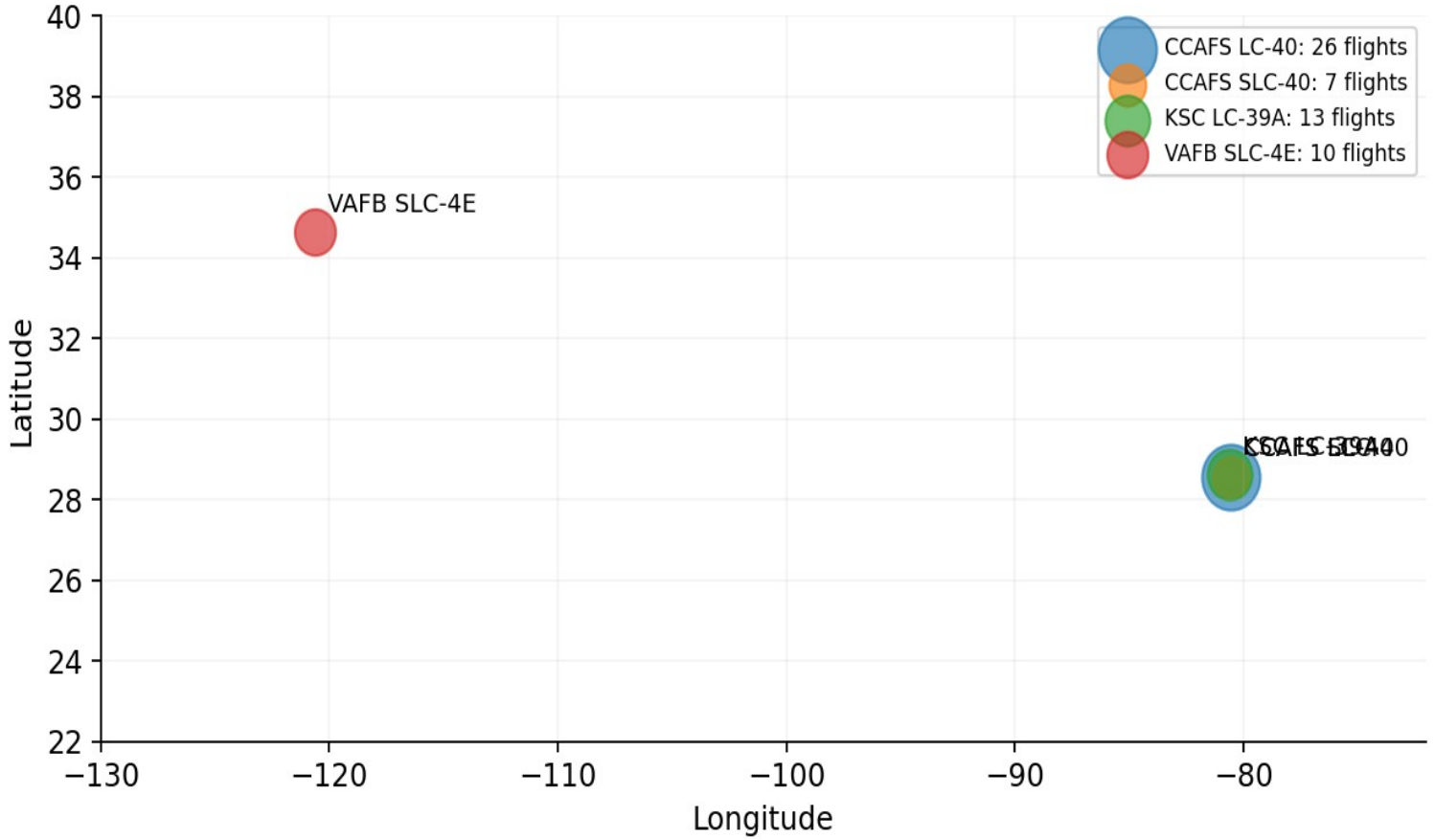
VAFB SLC-4E: 4/10 (40.0%)

CCAFS SLC-40: 3/7 (42.9%)

INTERPRETATION

Larger success counts may reflect a higher number of missions rather than a safer launch site. No causal site ranking is claimed.

Geospatial proximity | What is and is not measured



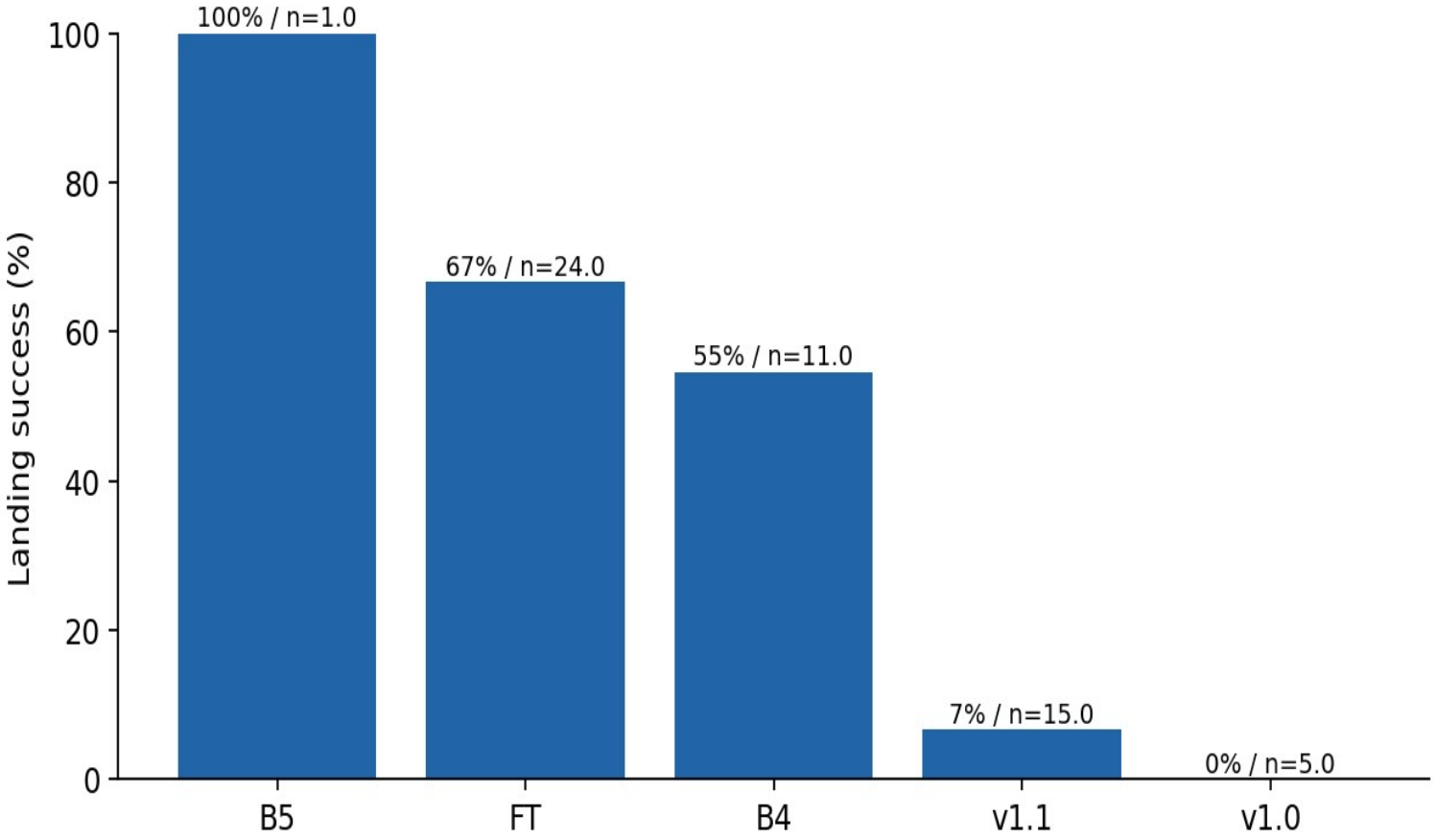
PROXIMITY CRITERION

The three launch coordinates can be mapped reliably. This historical CSV does NOT contain road, rail or coastline geometries, so kilometer distances to these features cannot be substantiated.

RESEARCH LIMITATION

To complete the infrastructure proximity criterion rigorously, geocode verified landmarks, calculate geodesic distances, then cite the map source. No invented distances are used.

Plotly Dash | All-sites recovery outcomes



DASHBOARD FUNCTION

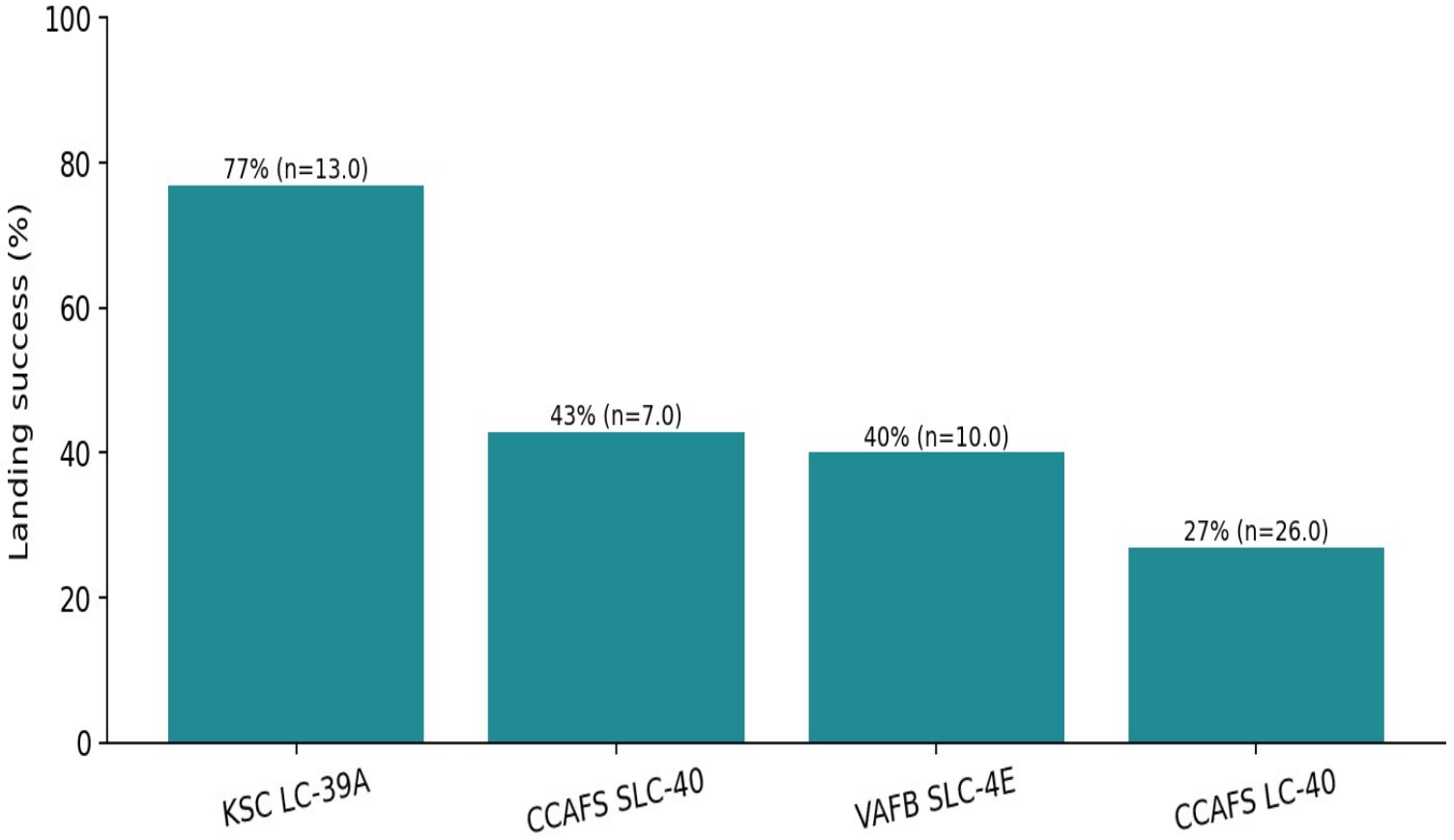
The companion dashboard_app.py implements a dropdown with an All Sites choice and individual pads, plus a payload range filter. The image on the left is an actual generated analysis export.

KEY METRIC

The full sample records 24 successful vs 32 other recoveries: 42.9% Class=1. This is booster recovery, not mission success.

Executable app bundled; this slide is not misrepresented as an actual browser screenshot.

Plotly Dash | Site-filtered comparison



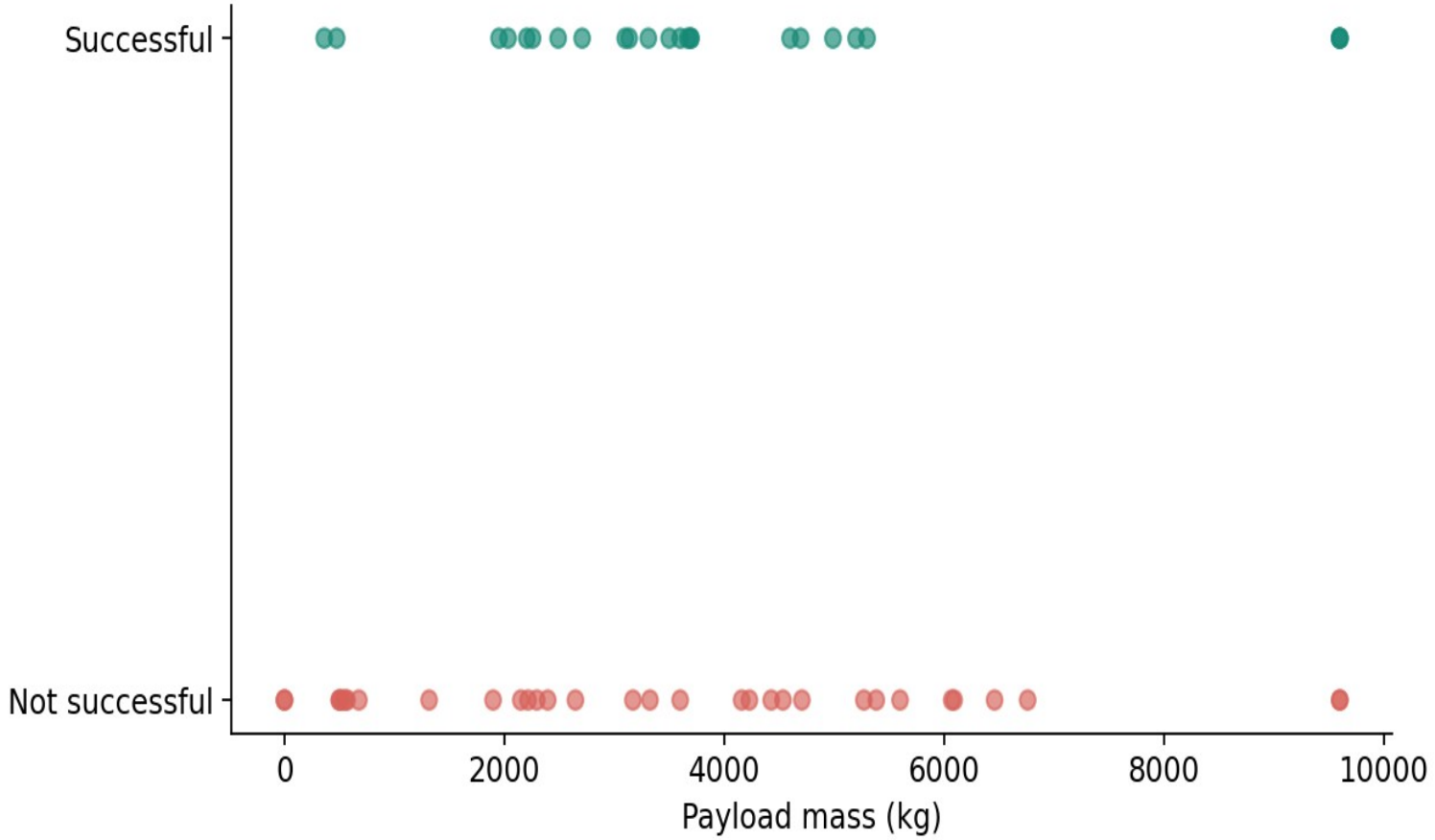
SITE DROPDOWN

Selecting a site changes the Dash figure to include only the chosen launch-pad records. The site-success chart documents the underlying aggregates and denominators.

DECISION-USEFUL READING

Use both the percentage and number of flights. A high recovery rate on very few launches is less informative than a stable rate across a larger sample.

Plotly Dash | Payload filter and outcome



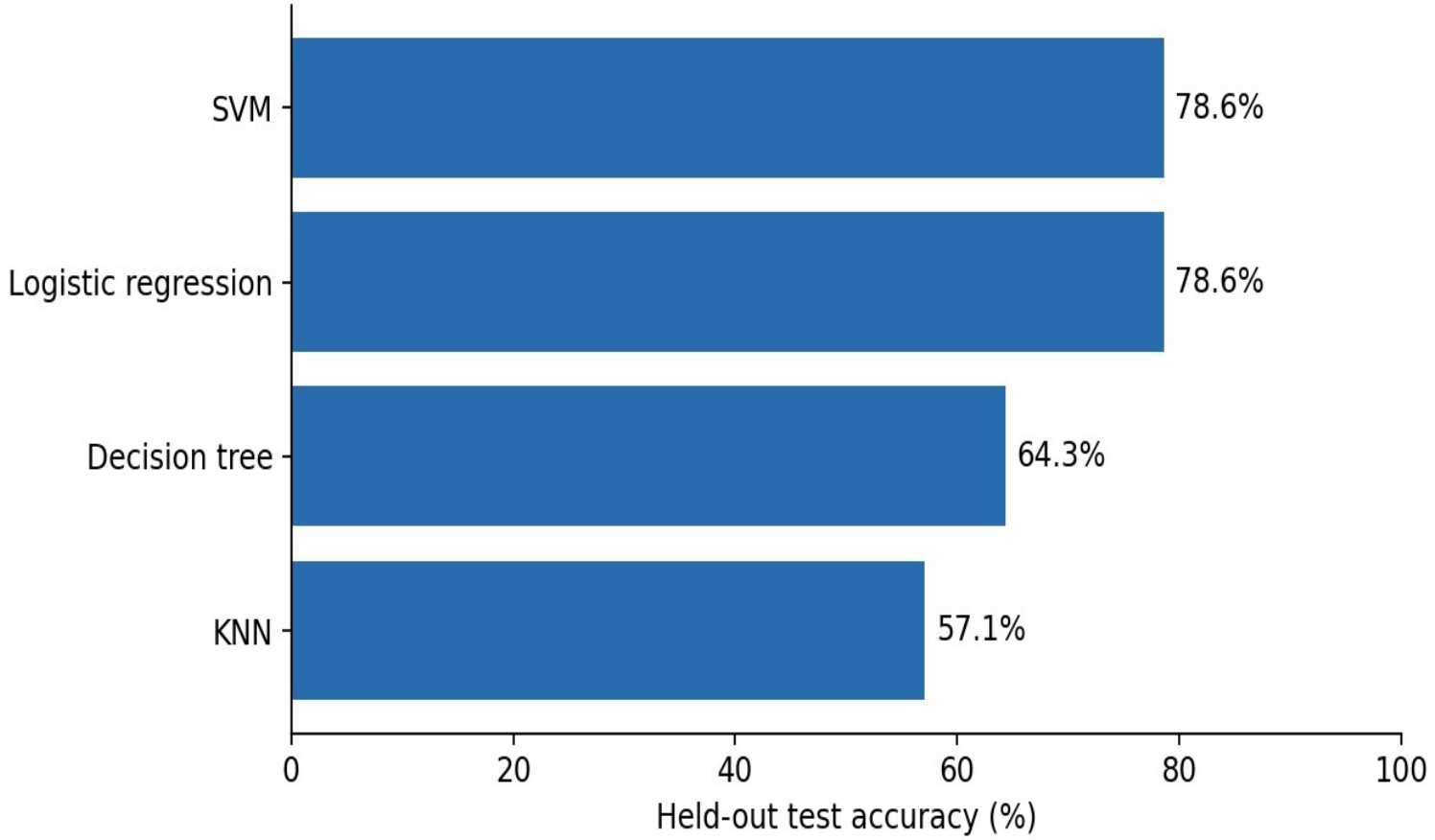
FILTER AND SCATTER

The Dash RangeSlider can restrict the observed payload interval. A scatter plot plots payload against flight number with color indicating recovery outcome.

INTERPRETATION

Apparent payload patterns can be entangled with flight order, hardware generation, and site. Filtering supports investigation, not proof of causation.

Predictive results | Why model selection matters



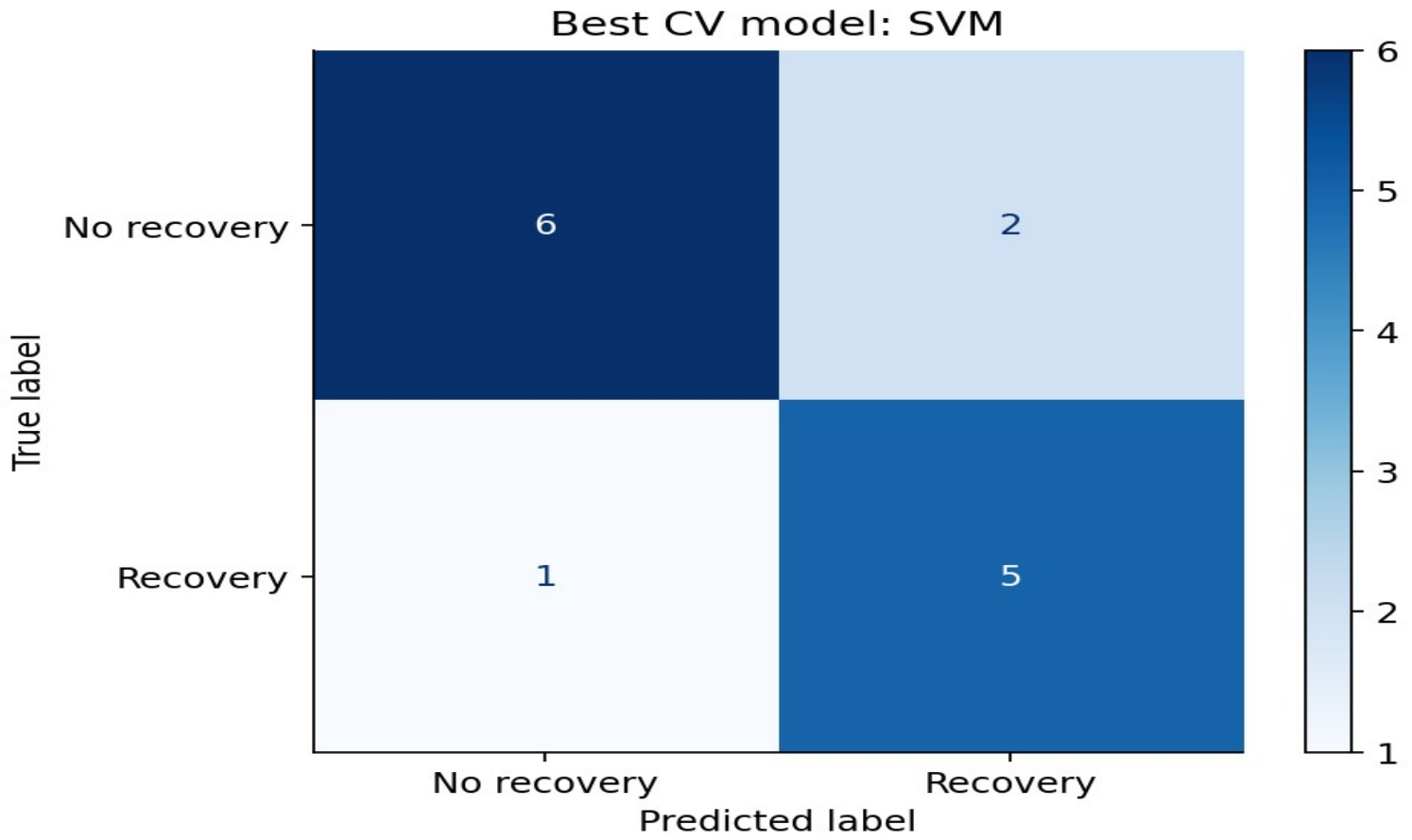
VALIDATION

SVM: 78.6% cross-validation; 78.6% holdout. Logistic regression: 73.8% CV, 78.6% holdout. Decision tree and KNN generalize less well on this split.

BEST MODEL RULE

SVM is selected by cross-validation, not post hoc by the holdout set. Test set includes only 14 records; small changes in counts materially affect the percentage.

Predictive results | Confusion matrix and errors



INTERPRETATION

Rows give observed recovery outcome; columns give prediction. Off-diagonal cases are errors. Accuracy alone obscures the class-specific consequences of mistakes.

LIMITATIONS

The model uses 42 training and 14 test records. A larger, newer dataset and time-based validation would be needed before deployment decisions.

Conclusions | Four evidence-linked takeaways

01 | Baseline

24 of 56 launches (42.9%) in this historical sample are labeled as successful first-stage recoveries. This number must not be generalized to present-day SpaceX operations.

02 | Explanatory patterns

Booster generation and flight progression vary with recovery outcome. Associations can reflect technological maturity, launch conditions, and mission composition.

03 | Model accuracy

SVM was strongest on cross-validation (78.6%) and correctly classified 11 of 14 held-out records. The small test set makes performance uncertain.

04 | Next decision

Use fresh and consistently labeled launch records, richer weather/trajectory features and chronological validation before comparing launch-provider reliability.

Reproducibility | Source links and evidence

Public course dataset	https://github.com/Ahmed-Nezar/IBM-Data-Science-Capstone-SpaceX/blob/main/spacex_launch_dash.csv
Interactive map	folium_site_map.html (included in the evidence package)
Dashboard code	dashboard_app.py (included in the evidence package)
SQL + classifier results	SpaceX_executed_analysis.ipynb, sql_results.txt, model_evaluation.csv
Personal GitHub URL	Not available: your repository must be published from your own account

Rubric audit | Outstanding before full credit

Done in report

PDF format; introduction; substantive executive summary; collection / cleaning / EDA methods; SQL results; model comparison and confusion matrix.

Included, but limited

Folium coordinate map, site success outcomes and source-backed proximity discussion; static outputs from a runnable Dash app.

Not verified complete

Actual browser-captured Folium and Dash screenshots; official IBM 90-launch notebook outputs; infrastructure distance calculations.

Must be supplied by learner

Personal public GitHub repository with completed original lab notebooks and Python files; a repository URL in the slides.